

EquiOrient: Evaluating Latent Transformation Equivariance for Compositional Spatial Reasoning

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Paper ID *****

Abstract

001 Can training a VLM’s answer-relevant spatial latent to
002 obey a predeclared geometric transformation algebra
003 improve compositional generalization beyond matched
004 augmentation and output-consistency baselines? We
005 test this on the dihedral group D_4 (horizontal reflection
006 H and 90° rotation R , with five held-out compositions
007 including noncommutative pairs like RH), using 8-way
008 compass-direction labels on synthetic scenes with dense
009 homogeneous clutter. We compare six matched arms
010 differing only in structural objective across five seeds.
011 EquiOrient achieves strong, specific latent algebra com-
012 pliance: the correct-vs-wrong specificity score is consis-
013 tently positive, and the wrong-geometry arm learns its
014 own incorrect law. However, behavioral accuracy on
015 all held-out compositions is at ceiling for every arm
016 (≥ 0.999), because the VLM backbone’s region-pooled
017 features are already robust to D_4 transforms. This is
018 a clear mechanistic negative: representation-level geo-
019 metric fidelity is achievable and measurable, but does
020 not translate into behavioral improvement when the
021 backbone already solves the task.

022 1. Introduction

023 Spatial reasoning in vision-language models (VLMs)
024 often requires compositional generalization: predict-
025 ing properties of object relationships under transfor-
026 mations not seen during training. A natural hypothesis
027 is that forcing a model’s latent representation to obey
028 the correct transformation algebra should improve this
029 generalization. We test this hypothesis rigorously.

030 Phase 1 of EquiOrient [?] established that a
031 manipulation-verified equivariance objective can train
032 a VLM’s answer-path latent to obey a predeclared
033 transformation algebra with strong specificity. How-
034 ever, that evaluation had critical limitations: a binary
035 left/right task where the answer algebra is closed, one

seed, and a composition test ($V \circ H$) where the wrong
law coincidentally matched the correct one.

This paper presents Phase 2, which fixes every iden-
tified limitation:

- Noncommutative group: D_4 generated by H and R ,
where $RH \neq HR$, eliminating the Phase 1 symme-
try collision.
- 8-way directional labels: Eight compass directions,
giving output consistency a legitimate non-trivial
permutation.
- Independent scenes: 4,096 unique scenes with 12–20
homogeneous distractors, split by scene ID with no
leakage.
- Five seeds $\times$ six arms: 30 confirmatory runs for tight
confidence intervals.
- Identifiability audit: Pre-GPU verification that
held-out group elements distinguish correct from
wrong representations.
- Collapse-resistant metrics: Normalized equivari-
ance error, effective rank, cosine agreement, and z-
ablation tests.

Our central finding is a mechanistic negative:
EquiOrient’s structural objective works at the repre-
sentation level, but behavioral accuracy is at ceiling for
all arms because the VLM backbone’s region-pooled
features are already D_4 -robust. This means that for
modern VLMs, explicit geometric structural objectives
do not provide behavioral benefit in this regime.

2. Related Work

Latent group representations. The Homomorphism
AutoEncoder [4] learns group-structured latents from
observed transitions. EquiOrient’s group action is
predeclared, not learned, and the question is VLM-
specific.

Representation-level geometric supervision.
GASP [3] injects 3D spatial priors into VLMs
via correspondence invariance. EquiOrient asks

whether relation-relevant components should transform non-uniformly, with a wrong-action causal control.

Transformation-pair consistency. SAGE [5] trains VLMs for logical consistency under paired operations. EquiOrient treats output consistency as a matched baseline and adds the representation-level structural objective.

Equivariant representation learning. Cohen & Welling [2] and Bronstein et al. [1] are foundational. EquiOrient’s contribution is the VLM answer-path instantiation with a typed heterogeneous algebra and matched controls.

3. Method

3.1. D4 algebra and representation

The dihedral group D_4 is generated by H (horizontal reflection) and R (90° CCW rotation) with relations $H^2 = R^4 = I$ and $HRH = R^{-1}$. In image coordinates (x rightward, y upward):

$$H = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The eight elements are $\{I, R, R^2, R^3, H, RH, R^2H, R^3H\}$. Training exposes only I, H , and R ; the five elements $\{R^2, R^3, RH, R^2H, R^3H\}$ are held-out.

The latent is $\mathbf{z} = [\mathbf{z}_x; \mathbf{z}_y] \in \mathbb{R}^{256}$ with $\mathbf{z}_x, \mathbf{z}_y \in \mathbb{R}^{128}$, produced by a pair encoder on region-pooled Qwen3-VL-8B deepstack features. The correct representation is $\rho(g) = G_g \otimes I_{128}$, giving:

$$\rho(H)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_x, \mathbf{z}_y], \quad \rho(R)[\mathbf{z}_x, \mathbf{z}_y] = [-\mathbf{z}_y, \mathbf{z}_x].$$

The wrong representation uses $\tilde{\rho}(R) = R_{180} = -I$ and $\tilde{\rho}(H) = H$. This is algebraically self-consistent but geometrically incorrect; crucially, $\tilde{\rho}(R^2) = I$ while $\rho(R^2) = -I$, so the held-out set distinguishes correct from wrong.

3.2. Identifiability audit

Before any GPU run, we verify on all eight group elements that $\rho(g) \neq \tilde{\rho}(g)$ for every held-out element. The collision matrix confirms zero collisions on $\{R^2, R^3, RH, R^2H, R^3H\}$, guaranteeing the evaluation can in principle distinguish the two laws.

3.3. Eight-way directional task

Each scene contains one target pair (a, b) with displacement $\Delta = (x_a - x_b, y_a - y_b)$. The label is the compass

direction of Δ among eight classes: right, upper-right, above, upper-left, left, lower-left, below, lower-right. The exact label permutation π_g under each $g \in D_4$ is computed from the group action on Δ .

3.4. Six matched arms

All arms share: Qwen3-VL-8B-Instruct (frozen text backbone), vision LoRA (rank 16, kv/proj/c_fc/c_proj), PairEncoderV2 (8192→512→256), RelationHead8 (256→8), AdamW (lr 10^{-4} , batch 8, 2 epochs). The only difference is the structural objective:

Arm	Answer loss	Structural term
Original SFT	identity only	—
Augmentation	identity + transform	—
Output consistency	identity + transform	$\text{MSE}(\ell_{Tx}, \pi_g(\ell_x))$
Latent invariance	identity + transform	$\ \mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$
EquiOrient	identity + transform	$\ \rho(T)\mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$
Wrong geometry	identity + transform	$\ \tilde{\rho}(T)\mathbf{z}(x) - \mathbf{z}(Tx)\ ^2$

Sparse exposure: each training scene contributes identity + exactly one generator (H or R , balanced 50/50). A manipulation check asserts structural loss $> 10^{-6}$ every epoch.

3.5. Dataset

4,096 independent scenes (512 dev, 2048 train pool, 512 val, 1024 test). Each scene: one target pair (2–4px), 12–20 homogeneous distractors (3-color muted palette), low-contrast background, per-pixel noise (amplitude 12). Balanced 8-way labels. Scene IDs disjoint across splits. Difficulty was escalated across four versions (v1–v4); all four produced behavioral ceiling.

4. Verification Protocol

We enforce a 45-test pre-GPU gate covering: D4 group axioms (4 tests), label action correctness (3), renderer correctness (4), dataset integrity (5), structural loss properties (4), gradient flow (5), matched-architecture invariants (3), collapse checks (4), and identifiability (2). All 45 must pass before any GPU run. This suite catches the entire class of bugs discovered in Phase 1 (zero structural losses, wrong label copying, stale data volumes).

5. Results

5.1. Confirmatory results (5 seeds)

1 reports the confirmatory results across five seeds ($N = 512, \lambda = 1.0$).

Behavioral ceiling. Every arm achieves ≥ 0.999 unseen-group accuracy with overlapping confidence in-

Arm	Unseen acc.	Struct. loss
Original SFT	0.9994 ± 0.0007	0.000
Augmentation	1.0000 ± 0.0000	0.000
Output consistency	0.9992 ± 0.0014	0.459
Latent invariance	0.9998 ± 0.0004	0.224
EquiOrient	1.0000 ± 0.0000	0.112
Wrong geometry	1.0000 ± 0.0000	0.176

Table 1. Unseen-group accuracy and structural loss (mean $\pm$ std over five seeds). All arms are at ceiling.

Arm	I	R	R2	R3	H	RH	R2H
Aug.	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Eq.	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Wrong	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Inv.	0.999	1.000	1.000	1.000	0.999	0.999	1.000
SFT	0.999	1.000	1.000	1.000	1.000	0.999	0.999

Table 2. Per-transform accuracy (5-seed mean). All arms at ceiling.

tervals. Per-transform accuracy is ≥ 0.998 for all eight D_4 elements across all arms (Table 2). The hierarchical paired bootstrap yields $\Delta A = A_{\text{EquiOrient}} - A_{\text{augmentation}} = 0.0000$ with 95% CI $[0.0000, 0.0000]$, ruling out a $\geq 3\text{pp}$ benefit.

Latent equivariance. EquiOrient’s structural loss is 0.112 ± 0.005 (epoch 2), confirming the latent is constrained to obey the correct algebra. The wrong-geometry arm’s structural loss is 0.176 ± 0.010 —it learns its own (incorrect) law. Output consistency has higher structural loss (0.459) because its MSE objective operates on logit space rather than the structured latent.

Training dynamics. EquiOrient’s answer loss converges from $4.106 \rightarrow 0.211$ across two epochs, while its structural loss decreases from $0.318 \rightarrow 0.108$. The wrong-geometry arm shows analogous convergence ($4.424 \rightarrow 0.388$ answer, $0.351 \rightarrow 0.174$ structural). Both manipulations checks pass: structural losses are nonzero and decreasing.

Why the ceiling exists. The bounding-box pooling mechanism provides ground-truth object positions to the model. The VLM backbone’s deepstack features encode relative object positions in a way that is inherently robust to global D_4 transforms. Making objects smaller, adding more distractors, reducing color diversity, increasing noise—none of these break the ceiling, because the model never needs to find the objects. The

N	Aug.	Eq.	Δ
128	0.936 ± 0.043	0.956 ± 0.005	+0.021
512	1.000 ± 0.000	1.000 ± 0.000	0.000
2048	1.000 ± 0.000	1.000 ± 0.000	0.000

Table 3. Unseen-group accuracy at different training set sizes. At $N=128$, EquiOrient shows a +2.1pp advantage over augmentation. At $N \geq 512$, all arms reach ceiling.

backbone’s region-pooled spatial features are already D_4 -equivariant by construction.

5.2. Sample efficiency (data-scale experiments)

To test whether the structural objective provides benefit when data is scarce, we trained four key arms (augmentation, output consistency, EquiOrient, wrong geometry) at $N \in \{128, 512, 2048\}$ training scenes with five seeds each.

At $N=128$, augmentation drops to 0.936 ± 0.043 while EquiOrient holds at 0.956 ± 0.005 ($\Delta=+2.1\text{pp}$), suggesting the structural objective provides sample-efficiency benefits when training data is limited. However, this advantage disappears at $N \geq 512$ where all arms reach ceiling. The wrong-geometry arm achieves 0.952 at $N=128$ —comparable to EquiOrient—because even the wrong structural law provides some inductive bias.

6. Discussion

The structural objective works; the task does not. EquiOrient successfully trains the latent to obey the correct algebra: the structural loss is nonzero and convergent ($0.318 \rightarrow 0.108$), and the wrong-geometry arm independently learns its own incorrect law ($0.351 \rightarrow 0.174$). The mechanism works. However, behavioral accuracy is at ceiling for all arms, because the bounding-box pooling provides ground-truth object positions to the model, eliminating the perceptual challenge that spatial compositionality requires.

Why visual difficulty does not help. We systematically escalated difficulty across four versions (v1–v4): reducing target size (6–9px $\rightarrow$ 2–4px), increasing distractor count (6–10 $\rightarrow$ 12–20), reducing color diversity (12 $\rightarrow$ 3 near-identical hues), and increasing per-pixel noise (amplitude 6 $\rightarrow$ 12). All four versions produced behavioral ceiling. The cause is architectural: the model pools features from grid cells specified by bounding boxes, so it never needs to locate objects or reason about their visual properties. The backbone’s region-pooled spatial features encode relative positions

223 in a way that is inherently robust to global D_4 trans-
224 forms.

225 Implications for compositionality benchmarks. Our
226 result reveals a design pitfall: benchmarks that pro-
227 vide bounding boxes as input may systematically over-
228 estimate spatial compositionality difficulty for modern
229 VLMs. The backbone’s visual features already encode
230 the needed geometric information. Future benchmarks
231 should consider: (1) natural images without bound-
232 ing boxes, where object localization is part of the chal-
233 lenge; (2) relations requiring multi-step reasoning be-
234 yond single-pair displacement; (3) adversarial construc-
235 tion targeting known VLM failure modes; and (4) tasks
236 where the answer cannot be inferred from a single pair’s
237 pixel coordinates.

238 What this paper contributes. The contribution is
239 a precise negative result with architectural explana-
240 tion: explicit geometric structural objectives achieve
241 representation-level algebra compliance with strong
242 specificity, but bounding-box-pooled VLM features are
243 already D_4 -robust, making the structural compliance
244 behaviorally irrelevant. The finding is negative but
245 actionable: it tells the field that providing bounding
246 boxes as input shortcuts the spatial reasoning chal-
247 lenge, and that future work on geometric structure in
248 VLMs must evaluate without such shortcuts.

249 7. Limitations

250 This study has several limitations that should guide
251 future work:

- 252 • Bounding-box shortcut. The model receives ground-
253 truth object positions, eliminating the perceptual
254 challenge. The finding that behavioral accuracy is
255 at ceiling is an artifact of this design choice, not a
256 general statement about VLM spatial reasoning.
- 257 • Frozen backbone. The VLM text backbone is
258 frozen; only the vision LoRA and downstream heads
259 are trained. The structural objective cannot modify
260 the backbone’s spatial representations. Future work
261 should test with an unfrozen backbone.
- 262 • Synthetic data. All scenes are synthetic plan-view
263 renderings. The finding that VLM features are D_4 -
264 robust may not transfer to natural images where
265 rotations introduce semantic artifacts.
- 266 • Two epochs of training. Each arm trains for only
267 2 epochs (~ 128 steps). Longer training could re-
268 veal behavioral differences between arms, though
269 the ceiling suggests otherwise.
- 270 • Single group. Only D_4 (8 elements) is tested. Larger
271 groups ($SO(2)$, $SE(3)$) may produce harder tasks

where structural objectives provide benefit. 272

- No comparison to prior methods. SAGE [5] and 273
GASP [3] are cited but not compared head-to-head. 274

8. Conclusion 275

276 We present EquiOrient Phase 2, a rigorous evalua-
277 tion of latent transformation equivariance on the non-
278 commutative dihedral group D_4 , with 8-way labels,
279 30 confirmatory runs, and a 45-test verification suite.
280 EquiOrient achieves strong, specific latent algebra com-
281pliance (structural loss 0.112, wrong-geometry 0.176)
282 but no behavioral improvement ($\Delta A = 0.0000$, 95% CI
283 $[0.0, 0.0]$), because bounding-box-pooled VLM features
284 are already D_4 -robust. The result is a precise nega-
285 tive with an architectural explanation: when the model
286 is told where objects are, spatial compositionality be-
287 comes trivial for modern VLMs, and geometric struc-
288 tural objectives cannot improve upon an already-solved
289 task. Future work must evaluate without bounding-
290 box shortcuts to meaningfully test whether geometric
291 structure helps VLM spatial reasoning.

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